# Nilesh Amarathunge

• 408/9, Pinwalapitiya Watte, Bamunawala, Kurunegala.
nileshamarathunge@gmail.com NileshAm Nilesh Amarathunge +94767706176
https://portfolio.nilesh.uk/
*Updated: March 14, 2026*

## Statement

I am a passionate and driven individual with a strong interest in machine learning, robotics and embedded systems. I am constantly seeking to expand my knowledge and skills through continuous learning, staying up to date with the latest advancements in technology to contribute to innovative and impactful solutions.

## Education

**University of Moratuwa** Moratuwa, Sri Lanka
*BSc.(Hons) in Electronics and Telecommunications Engineering.* *2024 - Present*
- CGPA : **3.8817**

**National Institute of Business Management.** Kurunegala, Sri Lanka
*Diploma in Software Engineering.* *2023 - 2024*
- Distinction pass with GPA **3.911**

**Maliyadeva College.** Kurunegala, Sri Lanka
*GCE Advanced Level in physical science stream* *2009 - 2023*
- Combined Mathematics : **A**, Physics : **A**, Chemistry : **A**, General English : **A**
- Z-score : **+2.2132**

## Projects

**Team N1K0 – Micromouse** 2025 – Present
*Micromouse Robotics Competition Project* ***Project link:*** *portfolio.nilesh.uk*
- Designed and manufactured a competitive micromouse for SLIIT Robofest 2025.
- Led full mechanical system design including chassis, drivetrain, and assembly.
- Designed a compact high-reduction (1:6) gear system for coreless motors.
- Performed full mechanical assembly and validation for smooth motor control.

**Analog Noise Cancelling Headphones** 2025 – 2026
*Analog Electronics Product Design* ***Project link:*** *portfolio.nilesh.uk*
- Led a team to design and prototype an analog noise-cancelling headphone system.
- Designed complete analog circuitry including noise reduction and volume control.
- Created PCB layouts using Altium Designer and assembled the full prototype.
- Tested and validated circuit performance under real-world conditions.

**Team Lynx – Battle Bot** 2025 – Present

*Combat Robotics Competition* ***Project link:*** *portfolio.nilesh.uk*

- Led a multidisciplinary team to design and manufacture a battle bot for RoboGames 2025.
- Designed a horizontal spinning blade weapon driven by a high-power DC motor.
- Completed full mechanical and structural design prior to manufacturing.
- Competed at RoboGames 2025 hosted by the University of Moratuwa.

**Falcon E-racing** 2025 – Present

*Formula Student Competition*

- Falcon E-racing is the Formula Student group established in University of Moratuwa
- Volunteering at Falcon E-racing group as an low voltage engineer.
- Designed and manufactured the PCB's and wired the battery cart which handles charging the battery.

**ThermoBand** 2025 – 2026

*Wearable Medical Device Product Design* ***Project link:*** *portfolio.nilesh.uk*

- Collaborated with electronics engineers to develop a consumer-ready wearable device.
- Designed enclosure optimized for comfort, durability, and manufacturability.
- Integrated sensors and electronics into final assembled product.
- Device enabled continuous body temperature monitoring and historical data tracking.

**E-Commerce Web Application** 2023 – 2024

*Full-Stack Web Development Project* ***GitHub:*** *Frontend, Backend*

- Led a team of three to design, implement, and deploy a full-stack e-commerce platform.
- Developed inventory management system for store employees.
- Implemented customer-facing features including browsing, ordering, and payments.

# Skills

### Languages

- Python (Proficient)
- C++ (Proficient)
- Java (Novice)

### Other

- SolidWorks (Proficient)
- Altium Designer (Proficient)

# Awards and Experience

**Robotic Minds Competition** 2015

*1st place in National level Robotics Competition.*

**National Level Inventors Competition** 2014 - 2015

*1st, 2nd and 3rd places in national and provincial level invention competitions.*

* "Sahasak Nimavum", held by SLIC, Ministry of Education, Sri Lanka.

**National Level Stamp Exhibition** 2017

*1st place in the National Stamp Exhibition organized by the Philatelic Bureau of Sri Lanka.*

## Extra Curricular

**Student Representative** 2026-Present

*Student Representative for Department of Electronic and Telecommunications Engineering*

**Member in Electronic club, university of Moratuwa.** 2025

*Web developer under the public relations branch of E-Club.*

**Board member in Philatelic Society.** 2021

*Vice Secretary of Philatelic Society, Maliyadeva College.*

**Board member in Inventors Club.** 2021

*Treasurer of Inventors Club, Maliyadeva College.*

**Scout** 2012 - 2020

*Active member of the Scout Troop at Maliyadeva College.*